Modeling Annotator Disagreement for Speech Emotion Attribute Prediction

1st Payal Saha
Indian Institute of Science
Bangalore, India
payalsaha@iisc.ac.in

2nd Sriram Ganapathy
Indian Institute of Science
Bangalore, India
sriramg@iisc.ac.in

***Abstract*—Speech emotion recognition (SER) is inherently subjective, and emotional labels often exhibit substantial disagreement across annotators. Conventional SER systems typically collapse multiple annotations into a single averaged target, thereby discarding potentially informative uncertainty and disagreement patterns. In this work, we investigate annotator-aware learning objectives for continuous speech emotion recognition using the MSP-Podcast corpus. Specifically, we study Deep Evidential Emotion Regression (DEER), which models annotator ratings using unimodal Gaussian uncertainty, and Mixture Density Networks (MDNs), which provide a more flexible framework for modeling potentially multi-modal annotation distributions in the Valence–Arousal–Dominance (VAD) space.**

To evaluate the effectiveness of uncertainty-aware learning, we apply these objectives to several large-scale speech foundation models, including Whisper, WavLM, and Qwen2-Audio. The learned representations are assessed using a SUPERB-style downstream evaluation protocol across several dimensional (VAD) and categorical emotion recognition benchmarks. We further analyze the distribution of annotator ratings in MSP-Podcast and show that while many utterances exhibit approximately unimodal annotation distributions, a substantial subset displays pronounced disagreement and non-Gaussian behavior, motivating more flexible distributional modeling approaches.

Experimental results demonstrate that uncertainty-aware objectives can effectively capture annotator variability and yield competitive transferable speech emotion representations. Our analysis further reveals complementary strengths of evidential and mixture-based uncertainty modeling across different foundation model architectures, providing insights into when each approach is most beneficial for subjective affective prediction.

***Index Terms*—speech emotion recognition, emotion representation learning, mixture density networks, uncertainty modeling, speech foundation models, annotator disagreement.**

I. Introduction

Speech emotion recognition (SER) plays an important role in affective computing, behavioral analysis, and human–computer interaction. A key challenge in SER is the inherent subjectivity of emotional perception, where the same utterance may receive different emotional interpretations from different annotators. Consequently, datasets such as MSP-Podcast provide multiple annotations per utterance, representing a distribution of perceived emotions rather than a single ground-truth label.

Most existing SER approaches simplify this problem by aggregating annotations into a single target value using averaging or majority voting. While effective, this process discards information about annotator disagreement and uncertainty, potentially limiting the quality and generalizability of the learned representations. This motivates annotator-aware learning approaches that explicitly model annotation distributions instead of relying solely on aggregated labels.

Recent uncertainty-aware methods such as Deep Evidential Emotion Regression (DEER) [1] have demonstrated the benefits of modeling annotation uncertainty. However, their interaction with modern speech foundation models remains largely unexplored. Furthermore, annotator distributions may not always follow unimodal Gaussian assumptions, motivating more flexible distributional approaches such as Mixture Density Networks (MDNs).

In this work, we investigate uncertainty-aware representation learning for continuous emotion prediction in the Valence–Arousal–Dominance (VAD) space. We compare three training objectives: (1) evidential regression using DEER, (2) mixture density networks (MDNs), and (3) conventional Concordance Correlation Coefficient (CCC)-based regression. These objectives are evaluated using multiple pretrained speech encoders, including Whisper, WavLM, and Qwen2-Audio.

To assess representation quality, we adopt a SUPERB-style downstream evaluation protocol spanning both dimensional and categorical emotion recognition tasks. We additionally analyze annotator distributions in MSP-Podcast to study the prevalence of Gaussian and non-Gaussian annotation behavior and its implications for uncertainty modeling.

The main contributions of this work are:

- We investigate uncertainty-aware emotion learning using evidential and mixture-density-based objectives on MSP-Podcast.
- We analyze annotator label distributions and study the prevalence of Gaussian and non-Gaussian annotation behavior.
- We evaluate uncertainty-aware supervision across multiple speech foundation models, including Whisper, WavLM, and Qwen2-Audio.
- We assess representation transferability through a SUPERB-style downstream evaluation protocol spanning dimensional and categorical emotion recognition tasks.

II. Related work

A. Speech Emotion Recognition

Speech emotion recognition (SER) has evolved significantly from traditional handcrafted acoustic feature pipelines toward large-scale self-supervised and pretrained speech representation learning. Earlier approaches relied on engineered descriptors such as MFCCs, pitch, energy, and prosodic statistics combined with statistical learning models. Recent SER research increasingly leverages large pretrained speech and speech-language models such as wav2vec 2.0, HuBERT, WavLM, Whisper, and Qwen2-Audio, which provide transferable speech representations across diverse downstream tasks.

Despite these advances, most SER systems treat emotion prediction as a point-estimation problem, typically using regression or classification objectives with a single aggregated label per utterance. This simplification ignores the inherent subjectivity in emotion perception and limits the ability of models to capture annotation variability.

B. Annotator-Aware and Uncertainty Modeling

A prominent uncertainty-aware approach for continuous speech emotion recognition is Deep Evidential Emotion Regression (DEER) [2]. DEER models emotion labels using a Normal-Inverse-Gamma (NIG) evidential distribution parameterized by $(\gamma, \nu, \alpha, \beta)$, where γ represents the predictive mean while the remaining parameters characterize predictive uncertainty. The framework optimizes

$$\mathcal{L}_{\text{DEER}} = \mathcal{L}_{\text{NLL}} + \lambda \mathcal{L}_{\text{reg}}, \tag{1}$$

where $\mathcal{L}_{\text{NLL}}$ is computed using the Student-t predictive distribution induced by the NIG prior and $\mathcal{L}_{\text{reg}}$ encourages consistency between predicted uncertainty and annotator variability.

However, evidential formulations assume a unimodal predictive distribution. In practice, emotion annotations may exhibit substantial disagreement and can deviate from Gaussian behavior. Such distributions may contain asymmetric, heavy-tailed, or potentially multi-modal structures that cannot be adequately represented by a single evidential distribution.

C. Distributional and Multi-Annotator Modeling

Beyond evidential approaches, mixture-based models such as Mixture Density Networks (MDNs) [3] have been proposed to capture multi-modal output distributions. From an annotator-aware perspective, different mixture components can potentially capture distinct groups of annotator opinions, making MDNs a natural candidate for modeling subjective emotional perception. MDNs represent the target as a weighted mixture of Gaussian components, enabling the modeling of complex and potentially multi-modal annotation patterns. Such approaches are particularly suitable for SER, where multiple annotators may express distinct but valid emotional interpretations.

D. Summary and Research Gap

Existing uncertainty-aware SER approaches are largely based on unimodal evidential formulations and have primarily been evaluated on source-dataset performance. Comparatively little attention has been given to understanding the statistical characteristics of annotator distributions, investigating whether annotation behavior is unimodal or multi-modal, and evaluating uncertainty-aware objectives using modern speech foundation models under transferable downstream evaluation protocols.

In this work, we investigate both evidential and mixture-based annotator-aware learning objectives for continuous speech emotion recognition. We further analyze the distributional properties of annotator ratings in MSP-Podcast and study how uncertainty-aware supervision influences representation transferability across multiple downstream dimensional and categorical emotion recognition tasks.

III. Datasets

A. MSP-Podcast

MSP-Podcast [7] serves as the primary training corpus in this work. The dataset contains approximately 100 hours of naturalistic speech collected from podcast recordings and provides continuous Valence–Arousal–Dominance (VAD) annotations obtained from multiple human annotators. The corpus contains 84,260 and 31,961 utterances in the official training and validation splits, respectively.

A key characteristic of MSP-Podcast is the availability of multiple annotator ratings per utterance, enabling the study of annotator disagreement and uncertainty-aware learning. Unlike acted emotion datasets, MSP-Podcast contains spontaneous and semi-natural emotional speech recorded under diverse acoustic conditions, making it particularly suitable for analyzing realistic emotional variability. Throughout this work, MSP-Podcast is used to train uncertainty-aware emotion representation models based on evidential regression, Mixture Density Networks (MDNs), and conventional CCC-based objectives.

B. Downstream Evaluation Datasets

To evaluate representation transferability, we perform downstream experiments on both dimensional emotion regression and categorical emotion classification benchmarks.

1) Regression Benchmarks: **MSP-IMPROV.** MSP-IMPROV [12] is a multimodal emotional interaction corpus containing approximately 9 hours of speech collected during dyadic interactions. The dataset includes both scripted and improvised emotional expressions and provides continuous VAD annotations. Following prior work, we perform speaker-independent evaluation using the standard session-based protocol.

WHISER. WHISER [13] is a naturalistic speech emotion corpus annotated with continuous Valence, Arousal, and Dominance labels. The dataset captures spontaneous emotional speech under real-world recording conditions and is used to evaluate the generalization of learned emotional representations in dimensional emotion recognition.

2) Classification Benchmarks: **IEMOCAP (4-class and 6-class).** IEMOCAP [8] contains approximately 12 hours of audiovisual dyadic interactions. We evaluate both the commonly used 4-class setup (angry, happy/excited, sad, neutral) and the 6-class setup (happy, sad, angry, neutral, excited, frustrated) using speaker-independent evaluation protocols.

MELD. MELD [9] is a multi-party conversational emotion recognition dataset derived from the television series *Friends*. The corpus contains over 13,000 utterances and supports a seven-class emotion recognition task.

CMU-MOSI. CMU-MOSI [10] consists of opinion videos annotated with sentiment intensity scores. Following common practice, we formulate the task as binary sentiment classification and use it to assess cross-domain transferability of learned speech representations.

RAVDESS. RAVDESS [11] is an acted emotional speech corpus recorded by professional actors. The dataset includes emotions such as neutral, calm, happy, sad, angry, fearful, disgust, and surprise, and is widely used for evaluating speech emotion recognition systems under controlled recording conditions.

C. Summary

MSP-Podcast is used exclusively for representation learning and uncertainty-aware training. The learned representations are subsequently evaluated on MSP-IMPROV and WHISER for dimensional emotion regression and on IEMOCAP, MELD, MOSI, and RAVDESS for categorical emotion recognition and sentiment classification tasks.

IV. Analysis of Annotator Distributions

The MSP-Podcast corpus provides multiple annotator ratings for each utterance in the Valence–Arousal–Dominance (VAD) space, making it well suited for studying annotator disagreement and uncertainty. Since uncertainty-aware methods such as evidential regression typically assume unimodal target distributions, it is important to understand the statistical characteristics of annotator responses before selecting an appropriate modeling framework.

In this section, we analyze the distribution of annotator ratings in MSP-Podcast. Specifically, we examine the number of annotations available per utterance and investigate the extent to which annotator ratings follow Gaussian-like distributions. The goal of this analysis is to assess whether unimodal uncertainty assumptions are sufficient for modeling annotator variability or whether more flexible distributional approaches, such as Mixture Density Networks (MDNs), may be required.

A. Annotation Distribution in MSP-Podcast

Figure 1 shows the distribution of annotator counts per utterance in MSP-Podcast. While most utterances contain annotations from five raters, a smaller subset contains considerably more annotations, enabling a detailed analysis of annotator variability and distributional characteristics.

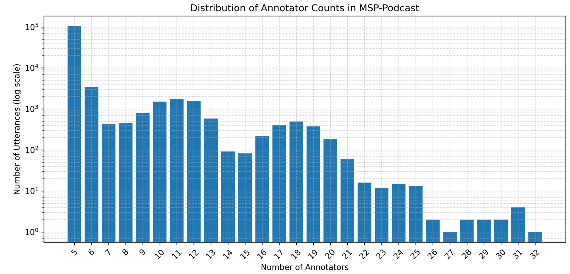


Fig. 1: Distribution of annotator counts per utterance in MSP-Podcast (log-scale y-axis). Most utterances contain five annotators, while a small subset contains considerably more annotations.

B. Gaussianity Analysis

To better understand annotator disagreement in MSP-Podcast, we analyze the distribution of annotator ratings for each utterance in the Valence–Arousal–Dominance (VAD) space. Since uncertainty-aware approaches such as evidential regression implicitly assume unimodal target distributions, this analysis aims to assess the validity of such assumptions.

For each utterance and emotion dimension, we perform the Shapiro–Wilk normality test on the available annotator ratings. Following standard practice, only utterances with at least four annotators are considered, and constant-valued annotations are excluded. An utterance is considered approximately Gaussian when the resulting p-value satisfies: $p > 0.05$

Our analysis shows that approximately 53.72%, 59.14%, and 52.19% of utterances exhibit Gaussian-like annotation distributions for Valence, Arousal, and Dominance, respectively. While this suggests that unimodal assumptions are often reasonable, nearly half of the utterances deviate from Gaussian behavior.

We additionally observe that many high-disagreement utterances exhibit asymmetric, broad, or otherwise non-Gaussian annotation distributions. These findings motivate the investigation of more flexible distributional models such as Mixture Density Networks (MDNs), which can represent complex and potentially multi-modal annotator distributions beyond unimodal evidential formulations.

V. Methodology

A. Framework Overview

Given an input utterance x, a pretrained speech encoder produces frame-level representations

$$H = h_1, h_2, \ldots, h_T. \tag{2}$$

These representations are aggregated using attentive statistics pooling to obtain an utterance-level embedding. Depending on the training objective, the pooled representation is passed to either an evidential regression head or a Mixture Density Network (MDN) head for predicting continuous Valence–Arousal–Dominance (VAD) annotations.

B. Speech Representation Extraction

We investigate multiple pretrained speech foundation models, including Whisper, WavLM, and Qwen2-Audio. For Whisper and WavLM, frame-level hidden representations are aggregated using attentive statistics pooling. Given attention weights α_t, the pooled mean and standard deviation are computed as:

$$\mu = \sum_{t=1}^{T} \alpha_t h_t, \tag{3}$$

$$\sigma = \sqrt{\sum_{t=1}^{T} \alpha_t h_t^2 - \mu^2} \tag{4}$$

where h_t denotes the frame-level representation and T is the number of temporal frames. The pooled mean and standard deviation are concatenated to form an utterance-level representation.

For Qwen2-Audio, we instead apply mean pooling over the frame-level hidden representations:

$$z = \frac{1}{T} \sum_{t=1}^{T} h_t, \tag{5}$$

which serves as the utterance-level representation for subsequent prediction heads.

The resulting utterance-level embeddings are used as inputs to the uncertainty modeling framework described in the following section.

C. Deep Evidential Emotion Regression

We adopt Deep Evidential Emotion Regression (DEER) [2] as our baseline uncertainty-aware objective. The training loss and evidential formulation remain identical to the original work. Our focus is on studying its interaction with different pretrained speech encoders and comparing it against Mixture Density Networks for annotator-aware emotion modeling.

D. Mixture Density Networks

While DEER assumes a unimodal predictive distribution, our Gaussianity analysis indicates that a substantial fraction of MSP-Podcast utterances exhibit non-Gaussian annotation behavior. To model such cases, we investigate Mixture Density Networks (MDNs).

Unlike DEER, which models annotator uncertainty using a unimodal predictive distribution, MDNs assume that annotator ratings are sampled from a conditional mixture distribution.

Given an utterance representation x, an annotator rating y is assumed to be generated according to

$$y \sim p(y \mid x) = \sum_{k=1}^{K} \pi_k(x) \, \mathcal{N}\left(y \mid \mu_k(x), \sigma_k^2(x)\right) \tag{6}$$

where, $\pi_k(x)$, $\mu_k(x)$, and $\sigma_k(x)$ denote the mixture weight, mean, and standard deviation of the k-th Gaussian component, respectively.

For each utterance, all available annotator ratings are treated as samples from the underlying annotation distribution. The MDN parameters are optimized by minimizing the negative log-likelihood:

$$\mathcal{L}_{\text{MDN}} = -\frac{1}{M} \sum_{m=1}^{M} \log \left(\sum_{k=1}^{K} \pi_k \, \mathcal{N}\left(y^{(m)} \,\middle|\, \mu_k, \sigma_k^2\right) \right) \tag{7}$$

where, $\{y^{(m)}\}_{m=1}^{M}$ denotes the set of annotator ratings for a given utterance.

The expectation of the conditional mixture distribution provides a point estimate of the emotion label:

$$\hat{y} = \mathbb{E}[y|x] = \sum_{k=1}^{K} \pi_k \mu_k. \tag{8}$$

This predictive mean is used as the final VAD prediction during evaluation.

E. CCC-based Regression

In addition to uncertainty-aware objectives, we consider a conventional regression baseline trained using the Concordance Correlation Coefficient (CCC) loss. For each utterance, multiple annotator ratings are first aggregated into a single target by averaging the corresponding VAD annotations. The model is then optimized to maximize the agreement between predictions and the aggregated targets using the standard CCC objective. Unlike DEER and MDN, this approach does not explicitly model annotator variability and serves as a point-estimate baseline for comparison.

F. Downstream Evaluation Protocol

To evaluate representation quality, we adopt a SUPERB-style downstream evaluation protocol. After training on MSP-Podcast, the encoder is frozen and used as a feature extractor for downstream regression and classification tasks. Layer-wise representations are combined using a learnable weighted sum and passed to lightweight task-specific heads. Regression tasks are evaluated using Concordance Correlation Coefficient (CCC), while classification tasks are evaluated using weighted accuracy and weighted F1-score.

VI. Experiments Setup

A. Training Data

All models are trained on the MSP-Podcast corpus using continuous Valence–Arousal–Dominance (VAD) annotations. Unlike conventional SER systems that use aggregated labels, we retain all available annotator ratings for each utterance to preserve annotation variability during training.

B. Speech Encoders

We investigate two pretrained speech foundation models: Whisper-Medium and WavLM-Large. For each encoder, we compare conventional CCC-based training, Deep Evidential Emotion Regression (DEER), and Mixture Density Networks (MDNs).

TABLE I: Downstream emotion recognition performance across datasets and encoders. Weighted F1-score (WF1, %) is reported for categorical emotion recognition datasets, and CCC is reported for emotion attribute prediction datasets. **Bold** indicates the best result

Dataset	Model	Pre-trained	CCC	MDN ($K = 2$)	MDN ($K = 3$)
Categorical Emotion Recognition					
IEMOCAP-4	WavLM-L	$71.7^{\pm 0.6}$	$73.6^{\pm 0.8}$	$74.1^{\pm 0.3}$	$\mathbf{75.2^{\pm 0.6}}$
	Whisper-M	$71.6^{\pm 1.0}$	$71.9^{\pm 0.3}$	$72.4^{\pm 1.1}$	$74.1^{\pm 1.2}$
	Qwen2-Audio-7B	$76.2^{\pm 0.8}$	$77.2^{\pm 0.4}$	$\mathbf{77.4^{\pm 0.3}}$	$77.0^{\pm 0.8}$
IEMOCAP-6	WavLM-L	$54.7^{\pm 0.9}$	$\mathbf{58.1^{\pm 0.5}}$	$57.6^{\pm 0.6}$	$58.0^{\pm 1.2}$
	Whisper-M	$55.7^{\pm 0.8}$	$55.9^{\pm 1.8}$	$57.2^{\pm 0.3}$	$\mathbf{57.5^{\pm 0.8}}$
	Qwen2-Audio-7B	$60.5^{\pm 0.5}$	$\mathbf{61.4^{\pm 0.4}}$	$60.7^{\pm 0.6}$	$61.1^{\pm 0.7}$
MELD	WavLM-L	$49.7^{\pm 0.3}$	$52.1^{\pm 0.9}$	$52.2^{\pm 0.4}$	$\mathbf{52.7^{\pm 0.3}}$
	Whisper-M	$50.7^{\pm 1.5}$	$51.9^{\pm 0.7}$	$\mathbf{52.8^{\pm 0.2}}$	$52.4^{\pm 0.3}$
	Qwen2-Audio-7B	$54.8^{\pm 0.3}$	$\mathbf{56.2^{\pm 0.3}}$	$55.7^{\pm 0.3}$	$55.2^{\pm 0.2}$
CMU-MOSI	WavLM-L	$68.9^{\pm 1.3}$	$\mathbf{77.4^{\pm 0.6}}$	$76.1^{\pm 1.1}$	$74.5^{\pm 0.9}$
	Whisper-M	$61.0^{\pm 0.5}$	$66.8^{\pm 2.4}$	$69.9^{\pm 1.9}$	$\mathbf{72.0^{\pm 1.9}}$
	Qwen2-Audio-7B	$73.6^{\pm 1.2}$	$\mathbf{83.8^{\pm 0.6}}$	$82.5^{\pm 0.7}$	$83.3^{\pm 0.7}$
RAVDESS-Song	WavLM-L	$63.2^{\pm 3.9}$	$63.1^{\pm 3.2}$	$\mathbf{66.5^{\pm 9.7}}$	$64.1^{\pm 9.2}$
	Whisper-M	$48.6^{\pm 2.7}$	$47.6^{\pm 4.4}$	$67.1^{\pm 7.8}$	$\mathbf{69.4^{\pm 7.5}}$
	Qwen2-Audio-7B	$87.6^{\pm 4.7}$	$87.2^{\pm 5.5}$	$\mathbf{88.5^{\pm 4.7}}$	$87.1^{\pm 5.5}$
Emotion Attribute Prediction					
MSP-IMPROV	WavLM-L	0.6220	0.6387	**0.6470**	0.6474
	Whisper-M	0.6494	**0.6569**	0.6562	0.6552
	Qwen2-Audio-7B	0.6227	0.6405	**0.6415**	0.6367
WHiSER	WavLM-L	0.5927	0.6249	**0.6513**	0.6512
	Whisper-M	0.6050	0.6293	0.6384	**0.6442**
	Qwen2-Audio-7B	0.5669	0.5905	**0.5975**	0.5967

C. Training Objectives

The following training strategies are evaluated:

- **Pre-trained**: frozen pretrained representations without MSP-Podcast fine-tuning.
- **CCC**: fine-tuning using Concordance Correlation Coefficient loss.
- **DEER**: fine-tuning using the evidential regression objective proposed in [2].
- **MDN**: fine-tuning using a mixture density network objective that models annotator ratings as samples from a conditional Gaussian mixture distribution.

For MDN experiments, we evaluate both $K = 2$ and $K = 3$ Gaussian mixture components.

D. Downstream Evaluation

Following a SUPERB-style evaluation protocol (as mentioned in previous section), on emotion recognition datasets (IEMOCAP-4, IEMOCAP-6, MELD, CMU-MOSI, and RAVDESS) and two dimensional emotion prediction datasets (MSP-IMPROV and WHISER).

Weighted F1-score (WF1) is reported for categorical tasks, while Concordance Correlation Coefficient (CCC) is reported for dimensional emotion prediction tasks.

VII. Results and Discussion

A. Downstream Emotion Recognition Performance

Table I summarizes the downstream evaluation results across five categorical emotion recognition datasets and two emotion attribute prediction datasets.

Overall, uncertainty-aware training consistently improves downstream performance compared to directly using pre-trained representations. Across categorical emotion recognition tasks, MDN-based objectives achieve the strongest average performance, with the $K = 3$ mixture model obtaining the highest overall average WF1-score. Similar trends are observed for dimensional emotion prediction tasks, where both MDN variants outperform conventional CCC-based training on average.

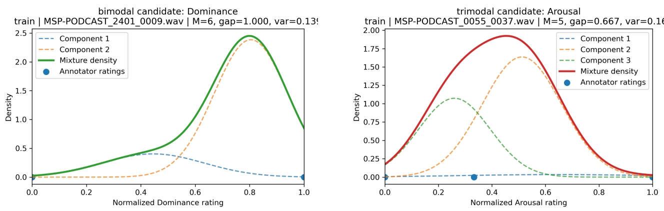
For WavLM-Large, MDN training yields consistent improvements across most downstream datasets, particularly on IEMOCAP and MELD. Whisper-Medium also benefits from mixture-based supervision, with the largest gains observed on CMU-MOSI and RAVDESS. Qwen2-Audio achieves the strongest overall downstream performance among the evaluated encoders and remains competitive under both CCC and MDN objectives.

These results suggest that preserving annotator distributions during training provides richer supervision than optimizing only aggregated emotion labels.

B. Effect of Mixture Components

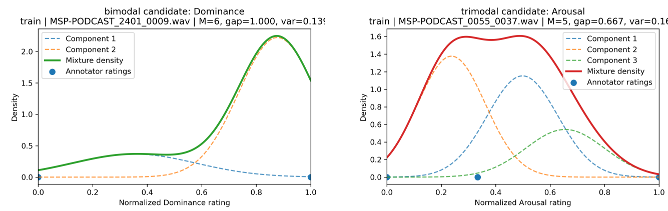
To investigate whether annotator distributions exhibit multimodal structure, we compare MDN models with $K = 2$ and $K = 3$ Gaussian components.

As shown in Table I, MDN-based training achieves competitive performance across a wide range of downstream emotion recognition tasks. While the relative performance of CCC and MDN objectives varies across datasets and encoders, the



(a) $K = 2$ (b) $K = 3$

Fig. 2: Representative annotation distributions learned by Whisper-based MDN models.



(a) $K = 2$ (b) $K = 3$

Fig. 3: Representative annotation distributions learned by WavLM-based MDN models.

MDN models frequently match or exceed the performance of conventional CCC-based training, indicating that modeling annotation distributions can provide a useful supervision signal beyond aggregated labels alone. While the performance differences between $K = 2$ and $K = 3$ are generally modest, the $K = 3$ models achieve slightly stronger average performance, suggesting that additional mixture flexibility can better capture annotation uncertainty.

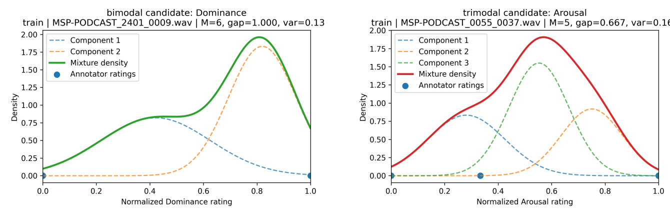
Figure 2, 3, 4 presents representative annotation distributions learned by Whisper, WavLM, and Qwen2-Audio using both $K = 2$ and $K = 3$ mixture configurations. For $K = 2$, the models capture bimodal annotation patterns corresponding to competing annotator interpretations of the same utterance. The learned Gaussian components are typically assigned to distinct regions of the label space, enabling the model to represent disagreement that cannot be adequately captured by a single Gaussian distribution.

Increasing the number of mixture components to $K = 3$ enables the models to represent more complex uncertainty patterns. In several high-disagreement utterances, the learned mixture components occupy distinct regions of the emotion space, producing multi-peaked or asymmetric predictive distributions. This behavior is particularly evident in the arousal examples, where the mixture components correspond to alternative low-, medium-, and high-arousal hypotheses. Among the evaluated encoders, Qwen2-Audio exhibits the clearest separation between mixture components, while WavLM and Whisper primarily use the additional component to increase distributional flexibility and capture broader uncertainty. Whisper-based model often produces comparatively flatter predictive distributions for K=3, suggesting that the additional mixture component is frequently used to model diffuse uncertainty rather than distinct annotation modes.

Overall, the qualitative examples support the hypothesis that a subset of emotional speech utterances exhibits annotation distributions that deviate from simple unimodal assumptions. Although the quantitative gains obtained by increasing the number of mixture components are relatively small, the learned distributions reveal richer annotator structures that are inaccessible to unimodal uncertainty formulations.

VIII. Conclusion

We presented a distribution-aware framework for emotion prediction based on Mixture Density Networks that explicitly



(a) $K = 2$ (b) $K = 3$

Fig. 4: Representative annotation distributions learned by Qwen2-Audio-based MDN models.

models annotator uncertainty. Across multiple downstream tasks, MDN-based supervision consistently outperformed conventional CCC-based training, demonstrating the benefit of learning from annotation distributions rather than aggregated labels alone. Qualitative analysis further revealed the presence of multi-modal annotation patterns that are not well represented by unimodal assumptions. These findings highlight the importance of uncertainty-aware learning for speech emotion representation learning.

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